

Current MCMC Set up

Code

```
true_PTPressure,true_errorNorm = generatingTrueValues(set_True_Cf,set_True_eta_Total,set_True_x_react)
prior_Cf = pm.TruncatedNormal("Cf", mu=set_Cf_Prior_mu, sigma=set_Cf_Prior_sigma,lower = 0,initval=set_Cf_Prior_mu,default_transform=None)
prior_eta_Total = pm.TruncatedNormal("eta_Total", mu=set_eta_Total_Prior_mu, sigma = set_eta_Total_Prior_sigma, lower = 0, upper = 1, initval=set_eta_Total_Prior_mu, default_transform=None)
prior_x_react = pm.TruncatedNormal("x_react", mu=set_x_react_Prior_mu, sigma = set_x_react_Prior_sigma, lower = 0, upper = 0.02, initval=set_x_react_Prior_mu, default_transform=None)

log_like = log_likelihood(prior_Cf,prior_eta_Total,prior_x_react,
                           pt.as_tensor_variable(true_PTPressure, dtype="float64"),
                           pt.as_tensor_variable(true_errorNorm, dtype="float64"),
                           pt.as_tensor_variable(errNorm_sigma, dtype="float64") )

pm.Potential("L1 Error Norm",log_like)

step = pm.DEMetropolisZ(
    vars = [prior_Cf,prior_eta_Total,prior_x_react],
    S= np.array([set_Cf_scale,set_eta_Total_scale,set_x_react_scale]),
    scaling = np.array([set_Cf_scaling,set_eta_Total_scaling,set_x_react_scaling]), #Initial scale factor for how aggressive the sampler noise moves around
    tune="scaling",
    tune_interval=100,
    tune_drop_fraction=0.9
)

trace = pm.sample(
    draws=set_draws,
    tune=set_tune,
    step = step,
    chains=set_chains,
    cores = set_cores,
    random_seed=42,
    progressbar=True,
    return_inferencedata=True,
    compute_convergence_checks=True,
)
```

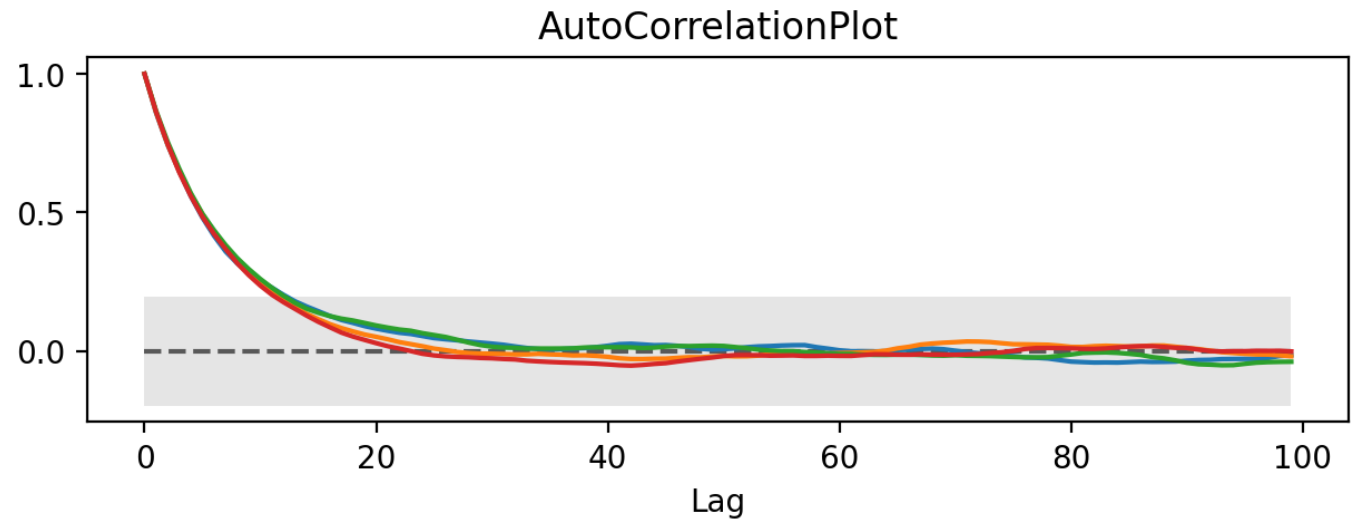
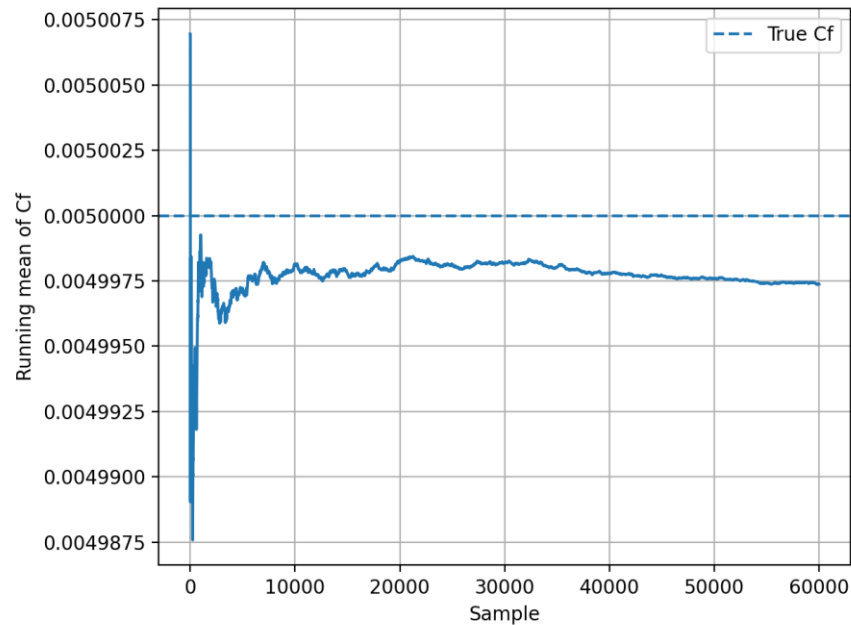
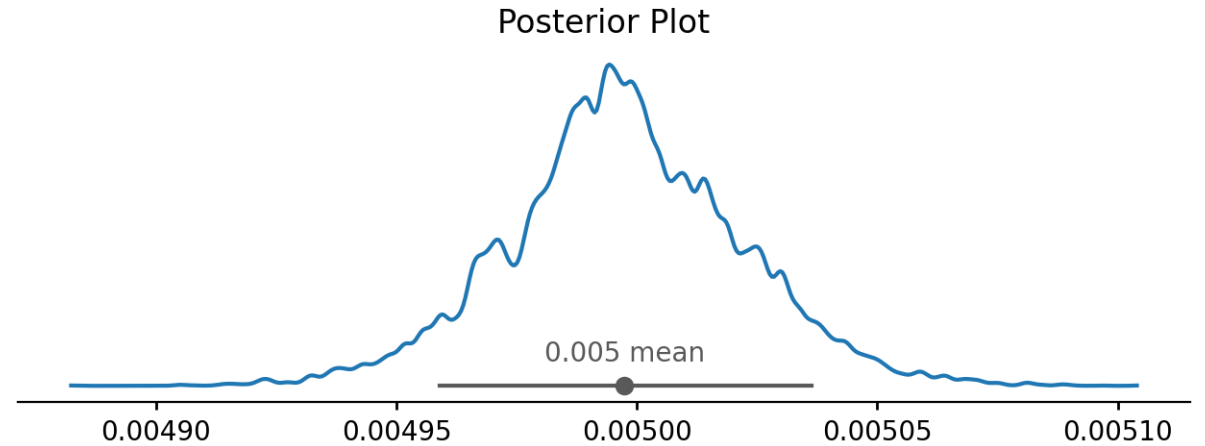
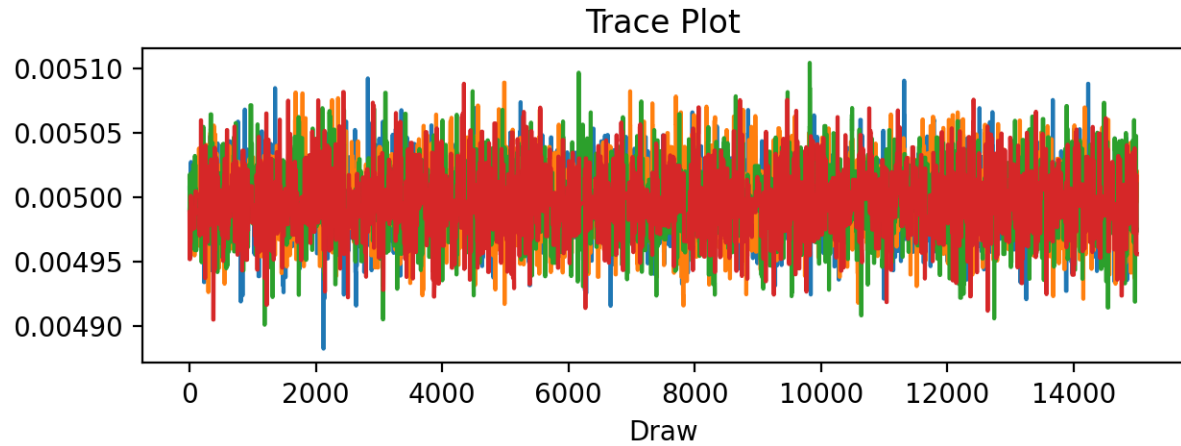
Likelihood Set up

- Work Flow:
 - Find bracket using input conditions from prior or priors.
 - Find scale using hybrid secant and bisection method
 - Resolve the forward model using that exact scale and get the static pressures at the PT locations
 - Get the residual PT values from parameters that are NOT true (predicted pressure values) – PT values from true parameter values (data pressure values)
 - Then take the L1 norm of those residuals and that's my input for log pdf
 - Likelihood mean = 0, std = will go over in later slides

1 Parmeter (Cf) – Successful Convergence

- Settings:
 - True CF = 0.005
 - Prior Mean = 0.00475
 - Prior Sigma = 0.00025
 - Likelihood Mean = 0.0
 - Likelihood Sigma = 2000.0
 - Draws = 15000
 - Tune = 2000
 - Chains = 4
 - Cores = 4
 - Scale = 0.001
 - Scaling = 0.5
- ARVIZ Summary
 - mean sd eti89_lb eti89_ub ess_bulk ess_tail r_hat mcse_mean mcse_sd
 - Cf 0.0049974 2.42e-05 0.005 0.005 4220 4293 1.00 3.7e-07 2.9e-07
- Results
 - Acceptance Rate: 0.1462
 - Posterior Mean :0.0049973716077675425
 - Posterior Median :0.004996594829263741
 - Posterior STD :2.4220973777144893e-05

1 Parmeter – Successful Convergence



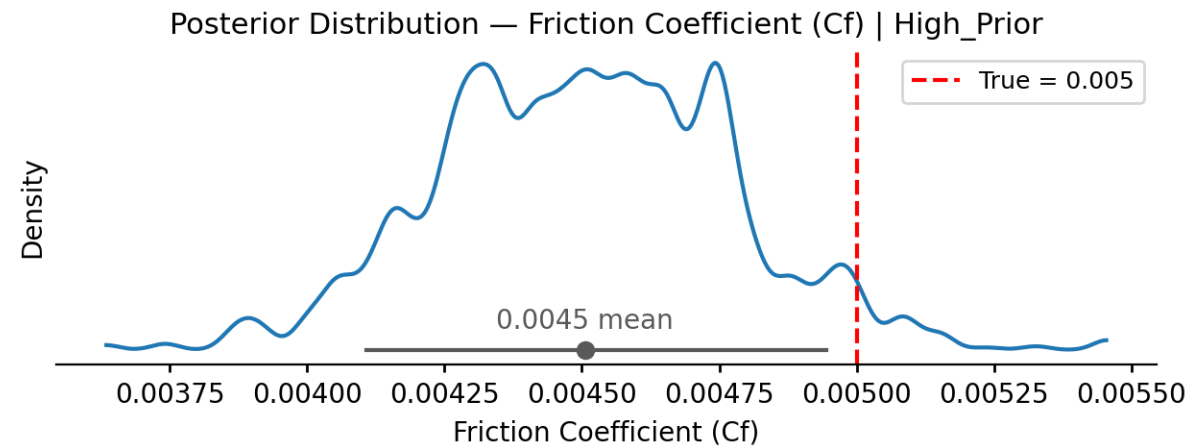
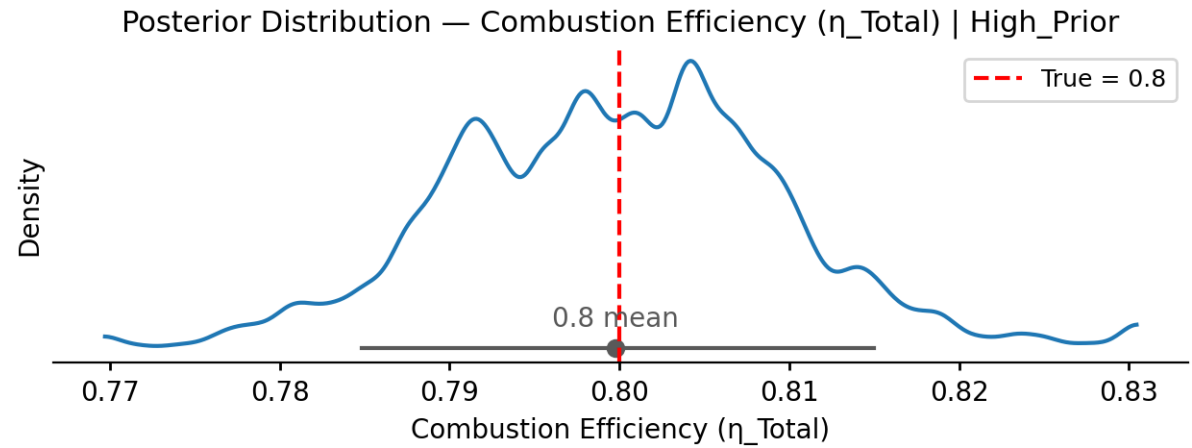
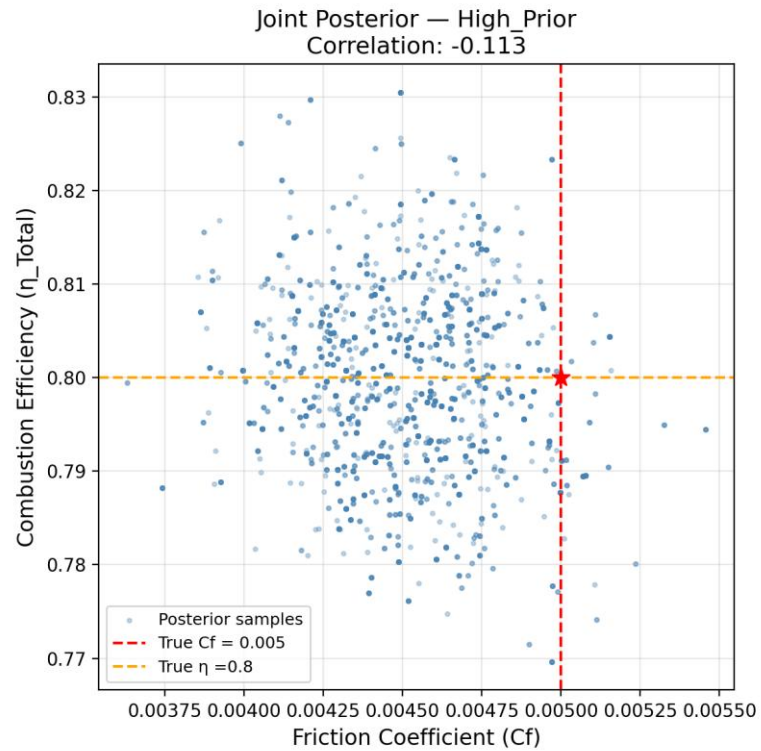
Conclusions for 1 param convergence

- Rhat and ESS are good so chain convergence/mixing is fine
- I tested this case with the exact same scaling but with different standard deviations and I would only ever recover on the true value if I had a small variance in the $1e3$ to $3e5$ range. If it was $1e4$ or high the chain would converge quite a bit under the true value
- I am assuming that this is because the changing the C_f does not have a large change on the pressure in the wind tunnel – even in the super sonic diverging section. The forward model output (PT Pressure) is not very responsive to even dramatic changes in C_f which results in the likelihood being somewhat flat

2 Parameter Cf and Eta Total

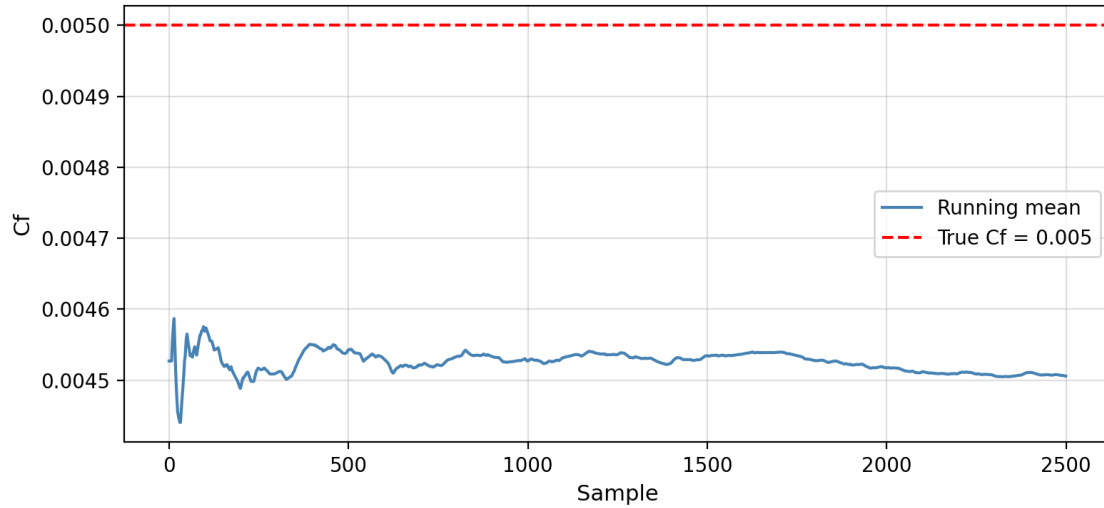
- Settings
 - True Cf = 0.005
 - Cf Prior Mean = 0.0045
 - Cf Prior Sigma = 0.00025
 - Cf Scale = 0.001
 - Cf Scaling = 0.01
 - True Eta = 0.8
 - Eta Total Prior Mean = 0.76
 - Eta Total Prior Sigma = 0.04000000000000001
 - Eta Total Scale = 0.1
 - Eta Total Scaling = 0.2
 - Likelihood Mean = 0.0
 - Likelihood Sigma = 100000.0Draws = 500
 - Tune = 500
 - Chains = 5
 - Cores = 5
- ARVIZ Summary
 - | | mean | sd | eti89_lb | eti89_ub | ess_bulk | ess_tail | r_hat | mcse_mean | mcse_sd |
|-----------|---------|---------|----------|----------|----------|----------|-------|-----------|---------|
| Cf | 0.00475 | 0.00025 | 0.0043 | 0.0052 | 247 | 235 | 1.02 | 1.6e-05 | 1.1e-05 |
| eta_Total | 0.799 | 0.01 | 0.78 | 0.81 | 145 | 249 | 1.03 | 0.0008 | 0.00055 |
- Other Results and metrics
 - Acceptance Rate: 0.156
 - Cf-Eta Total, Correlation Coeff: -0.12512375940286158

2 Parameter Cf and Eta Total

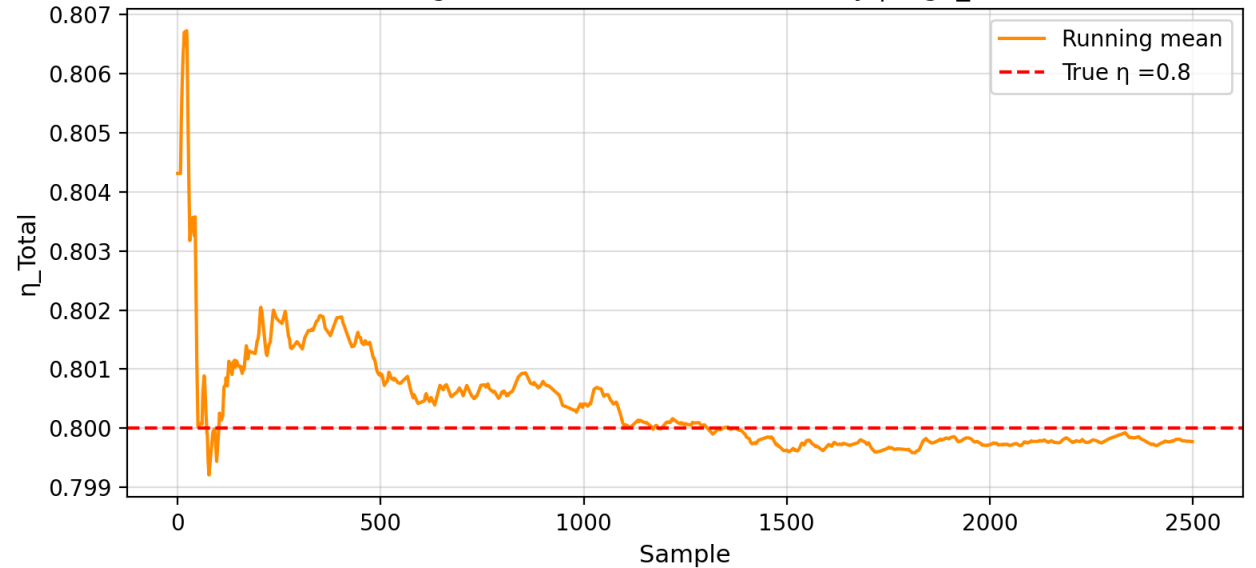


2 Parameter Cf and Eta Total

Running Mean — Friction Coefficient | High_Prior



Running Mean — Combustion Efficiency | High_Prior



Conclusions About 2 Parameter Stuff

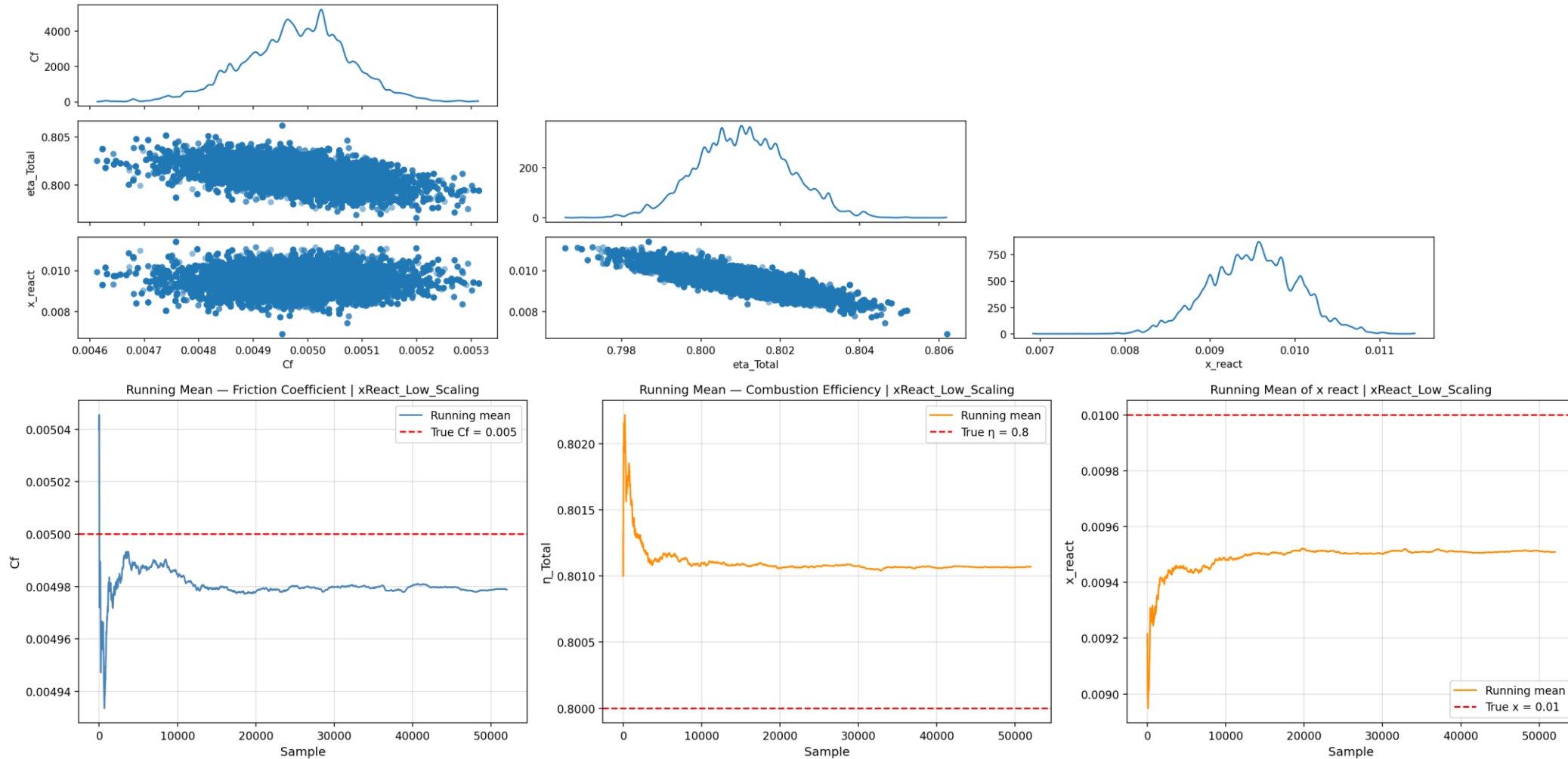
- I realized that my likelihood variance was too big, which meant that it did not have a strong effect on the posterior.
- I think that η_{total} converges and C_f does not because because the forward solver is way more sensitive to changes in η_{total} than changes in C_f , so η sort of dominates the recovery

3 Parameter Cf, Eta Total and x react

- Settings:
 - True CF = 0.005
 - Cf Prior Mean = 0.00475
 - Cf Prior Sigma = 0.00025
 - Cf Scale = 0.001
 - Cf Scaling = 0.1
 - True Eta Total = 0.8
 - Eta Total Prior Mean = 0.76
 - Eta Total Prior Sigma = 0.04000000000000001
 - Eta Total Scale = 0.1
 - Eta Total Scaling = 0.2
 - True x react = 0.01
 - x react Prior Mean = 0.0095
 - x react Prior Sigma = 0.0005
 - x react Scale = 0.01
 - x react Scaling = 0.1
 - Likelihood Mean = 0.0
 - Likelihood Sigma = 2000.0
- ARVIZ Summary
 - | | mean | sd | eti89_lb | eti89_ub | ess_bulk | ess_tail | r_hat | mcse_mean | mcse_sd |
|-----------|----------|---------|----------|----------|----------|----------|-------|-----------|---------|
| Cf | 0.004979 | 9.6e-05 | 0.0048 | 0.0051 | 925 | 924 | 1.01 | 3.2e-06 | 2.4e-06 |
| eta_Total | 0.80107 | 0.00115 | 0.8 | 0.8 | 1258 | 1309 | 1.01 | 3.3e-05 | 2.3e-05 |
| x_react | 0.00951 | 0.00052 | 0.0087 | 0.01 | 1038 | 983 | 1.01 | 1.6e-05 | 1.1e-05 |
- Other Results and metrics
 - Acceptance Rate: 0.09321153846153846
 - Settings:
 - Draws = 6500
 - Tune = 500
 - Chains = 8
 - Cores = 8

3 Parameter Cf, Eta Total and x react

Joint Posterior



Conclusions about 3 parameter stuff

- Pretty much the same as with the 2 parameter stuff but it seems that with a smaller likelihood variance I get better recovery on for low sensitivity variables like Cf. What im not sure about though is why my x react is not recovering even at such a low likelihood variance
- Running another case with lower variance to see if it helps